Sustainable Product / Lifestyle Recommender System

A Group Project Report for the
Course Recommendation Systems (DS2803)

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Abstract

This project presents a **Sustainable Lifestyle and Product Recommendation System** designed to promote environmentally responsible behavior by providing personalized recommendations based on individual lifestyle patterns. The system utilizes survey-based data capturing various aspects of daily life, including diet, transportation, energy consumption, and purchasing habits.

Through exploratory data analysis (EDA), key factors influencing sustainability ratings are identified, such as energy source, transportation mode, and consumption patterns. Users are profiled based on their sustainability score, enabling segmentation into different sustainability levels.

A hybrid recommendation approach is implemented, combining rule-based logic with data-driven insights to generate actionable suggestions for improving sustainability. Additionally, machine learning techniques such as clustering are explored to group users with similar behavior patterns and enhance recommendation quality.

The final system aims to provide personalized, practical, and impactful recommendations that encourage users to adopt more sustainable habits, contributing to environmental conservation and responsible consumption.

Keywords

Sustainable Recommender System, Lifestyle Analysis, Sustainability Rating, Personalized Recommendations, Machine Learning, Clustering, Environmental Awareness, Green Consumption.

Introduction

With increasing concerns about climate change and environmental degradation, promoting sustainable living has become a global priority. Individual lifestyle choices—such as transportation methods, dietary habits, and energy consumption—play a significant role in determining one’s environmental impact.

Despite growing awareness, many individuals struggle to identify actionable steps toward sustainability due to a lack of personalized guidance. Existing solutions often provide generic advice that does not account for individual habits, preferences, or constraints, limiting their effectiveness.

This project addresses this gap by developing a **Sustainable Lifestyle and Product Recommender System** that leverages data-driven insights to provide tailored recommendations. By analyzing user-specific lifestyle data, the system identifies patterns and key factors influencing sustainability ratings.

The proposed system integrates exploratory data analysis, user profiling, and recommendation techniques to generate meaningful suggestions aimed at improving sustainability levels. By offering personalized and actionable recommendations, the system encourages users to gradually transition toward more environmentally responsible behaviors.

Objectives

1. To analyze lifestyle and behavioral data to identify key factors affecting sustainability ratings.
2. To develop a user profiling mechanism that categorizes individuals into different sustainability levels based on their lifestyle patterns.
3. To design a personalized recommendation system that suggests eco-friendly alternatives in areas such as transportation, diet and consumption habits.
4. To apply machine learning techniques (e.g., clustering) to group users with similar sustainability behaviors and improve recommendation accuracy.
5. To build a system that provides actionable, easy-to-understand recommendations to encourage sustainable living.

Proposed Solution

The proposed system is designed as a **data-driven recommendation framework** that analyzes user lifestyle data and generates personalized sustainability recommendations through multiple processing stages.

Progress to date

- **Dataset Sourcing and Preprocessing:**
  Collected a relevant lifestyle and Product datasets and performed data cleaning, handling missing values and encoding features for analysis.
- **Exploratory Data Analysis (EDA):**
  Conducted EDA to identify key patterns and factors affecting sustainability, such as transportation, energy usage and consumption behavior.

- **Recommendation Engine Development:**
  Built an initial hybrid recommendation system combining rule-based logic with data-driven insights to generate personalized suggestions.
- **Prototype UI:**
  Developed a basic user interface to input user data and display sustainability recommendations in a simple and interactive manner.